

HAORAN JIN

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EDUCATION

University of Michigan

Sep. 2023 - present

Ph.D Candidate in Computer Engineering | Computer Architecture & EDA

Advisor: Prof. Nathaniel Bleier & Previous Advisor: Prof. George Tzimpragos

Shanghai Jiao Tong University

Sep. 2019 - Aug. 2023

B.Eng in Electrical and Computer Engineering

HONORS AND AWARDS

- 2026 SIGARCH ISCA Conference Student Travel Grants
- 2026 Rackham Conference Travel Grant
- 2019,2020,2021,2022 John Wu & Jane Sun Sunshine Scholarship
- 2020,2021,2022 SJTU Undergraduate Excellence Scholarship

PUBLICATION

- **Haoran Jin***, Kangqi Zhang*, Jirong Yang, Barry Lyu, Qiuyi Ding, Ruijie Gao, Nathan Bleier, *Precision As You Need: Stochastic Computing Is a Dense Adaptive Quantizer*, NeurIPS 2026 (Under Review)
- Yiyu Sun, Xinyang Han, ..., **Haoran Jin**, et al. *Agents' Last Exam*, arXiv 2026
- Ruijie Gao, **Haoran Jin**, Jirong Yang, Nathaniel Bleier, *TensorLift: Automatic Extraction of Tensor-Level ISA Semantics from Accelerator RTL via MLIR Semantic Lifting*, ICCAD 2026
- **Haoran Jin**, Jirong Yang, Zhiheng Zhang, Justin Shin, Barry Lyu, Kangqi Zhang, Yunpeng Liu, Nathaniel Bleier, *Fengshui: Demystifying Chiplet Ecosystem and Bespoke Neural Network Accelerator Codesign*, MICRO 2026
- **Haoran Jin***, Jirong Yang*, Barry Lyu, Ruijie Gao, Nathaniel Bleier, *ϵ SIP: μ Arch-aware ASIP-ISA Co-Design via Program Synthesis, Equality Saturation, and External Don't Cares*, ISCA 2026
- Panagiotis Papanikolaou, Alex Vanasse, **Haoran Jin**, Georgios Tzimpragos, Jennifer Volk, *Duplication-Aware Retiming and Cell Interface Redesign for Superconductor Circuit Minimization*, DAC 2026
- Zexi Li*, **Haoran Jin***, Kuncai Zhong, Guojie Luo, Runsheng Wang, Weikang Qian, *SCGen: A Versatile Generator Framework for Agile Development of Stochastic Circuits*, DATE 2024
- Kuncai Zhong, Zexi Li, **Haoran Jin**, Weikang Qian, *Exploiting Uniform Spatial Distribution to Design Efficient Random Number Source for Stochastic Computing*, ICCAD 2022

WORK EXPERIENCE

Intern, Architecture at *Tenstorrent*

May. 2026 - present

- Build performance model and do performance correlation for AI hardware

Graduate Student Instructor at *University of Michigan*

Jan. 2026 – May. 2026

- Organized the course, EECS 598 Computer Architecture for Post-Moore Era

SELECTED PROJECTS

ϵ SIP: μ Arch-aware ASIP-ISA Co-Design via Program Synthesis, Equality Saturation, and External Don't Cares

Prof. Nathaniel Bleier at *University of Michigan*

July. 2025 – Nov. 2025

- Proposed æSIP, the first µArch-aware ASIP-ISA co-design framework that leverages hardware aware program rewriting to automatically generate an optimized ASIP-program pair.
- Built a program synthesis framework that leverages neuro-symbolic approach to automatically discover correctness-preserving rewrites that maximize hardware removal opportunities.

Mozart: An Ecosystem-Accelerator Codesign Framework for Composable, Heterogeneous Chiplet-based Neural Network Accelerators

Prof. Nathaniel Bleier at *University of Michigan*

Jan. 2025 – Aug. 2025

- Developed a chiplet ecosystem-accelerator codesign framework to increase chiplet reuse and hence reducing non-recurring engineering costs.
- Built hierarchical design space exploration methodology to systematically compose heterogeneous chiplet-based bespoke ASIC leveraging operator-level disaggregation insights.

Minimum Area Retiming for xSFQ logic

Prof. George Tzimpragos at *University of Michigan*

Aug. 2024 – Dec. 2024

- Developed a linear programming formulation to simultaneously minimize the number of registers and duplication logic in xSFQ circuits for optimal area reduction while maintaining clock period constraints.

Mixed Circuit of Binary & Stochastic Computing

Prof. Weikang Qian at *SJTU*

Jan. 2022 – Sep. 2023

- Defined a domain-specific language (DSL) for agile mixed circuit development.
- Evaluated mixed binary-stochastic circuits for image processing applications.

SKILL

Programing Tools

C, C++, Python, Java, Rust, Verilog, Chisel, ELM, Racket
Triton, Timeloop, MatLab, Mathematica, Vitis HLS, Spark, Egglog, Hadoop, Pin, Pspice